BASELINE survey

Q1.1 **Welcome to the research study!** Please create your personal ID by combining the first 2 letters of your surname and the day of your birth (2 digits). Example: if your surname is Hansen and you were born on the 7th, your ID is HA07

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Q2.1 **Introduction:**
This survey is about the possible use of predictive models to help the university identify and contact students who may be experiencing academic difficulties, so that support can be offered early. **There are no right or wrong answers.** We are interested in your perspective on how such a system should be designed and used, and under what conditions it would be acceptable.

In this study, predictive models are considered as tools to help the university identify **students who may be experiencing academic difficulties**, so that support can be offered proactively.

The model would be trained using patterns from past student outcomes, such as dropout, because these are among the few outcomes the university can reliably observe. However, the purpose of the system is not only to predict dropout, but more broadly to support earlier and more targeted outreach to students who may be experiencing academic difficulties.

Throughout the survey, please assume that any data used follows applicable privacy laws and institutional safeguards, and that the model is used only to offer support to students.

Q3.1 **Data and Privacy:**
Imagine the university uses a predictive model to identify students who may be experiencing academic difficulties so it can offer support early.

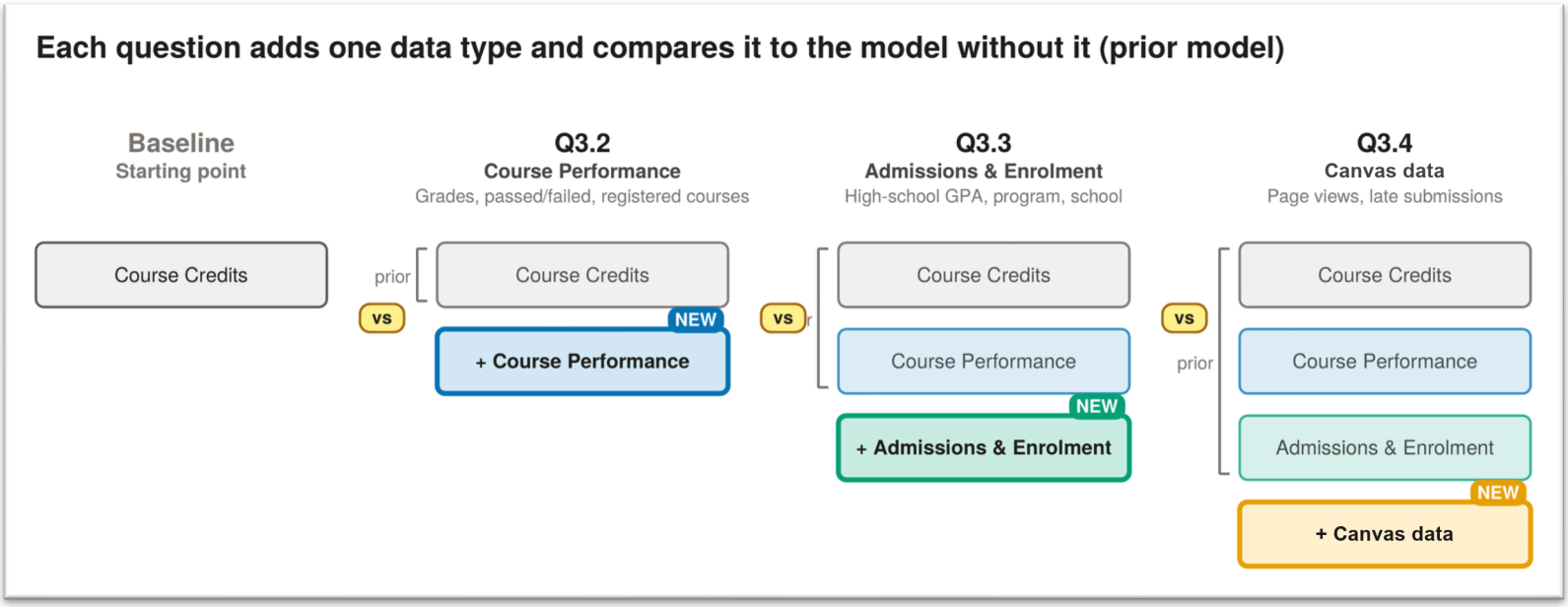
The model begins with course credits earned. The university could add other data sources that may improve identification, but may raise privacy concerns.

There are competing perspectives:

- Some argue that we should use any data that helps identify students experiencing academic difficulties - not using available data means missing students who need help.
- Others argue that some data is too invasive - surveillance risks and privacy violations can outweigh the benefit of identifying more students experiencing academic difficulties.

The figure below shows how the model grows across the following questions. For each question, **consider whether the highlighted addition is acceptable.**

In the questions below, an **improvement** means: the model correctly identifies more students experiencing academic difficulties out of 100, while the number of students incorrectly flagged stays about the same.



Q3.2 Would it be acceptable for the model to additionally use **course performance data** (such as grades, passed or failed courses, or number of registered courses) in addition to a model that uses only the number of course credits earned?

- ○ Acceptable even if it does not improve identification
- ○ Acceptable only if there is at least a small improvement (about 1–3 additional students identified out of 100 students experiencing academic difficulties)
- ○ Acceptable only if there is at least a moderate improvement (about 5 additional students identified out of 100 students experiencing academic difficulties)
- ○ Acceptable only if there is at least a large improvement (about 10 additional students identified out of 100 students experiencing academic difficulties)
- ○ Never acceptable

Q3.3 Would it be acceptable for the model to incorporate **administrative admissions and enrolment data** (e.g., high-school GPA (Grade point average), program, school/college), in addition to course credits earned and course performance metrics?

- ○ Acceptable even if it does not improve identification
- ○ Acceptable only if there is at least a small improvement (about 1–3 additional students identified out of 100 students experiencing academic difficulties)
- ○ Acceptable only if there is at least a moderate improvement (about 5 additional students identified out of 100 students experiencing academic difficulties)
- ○ Acceptable only if there is at least a large improvement (about 10 additional students identified out of 100 students experiencing academic difficulties)
- ○ Never acceptable

Q3.4 Would it be acceptable for the model to also use **online course activity data from the university learning platform Canvas** (for example, how often course pages are accessed or whether assignments are submitted late) in addition to a model that uses course credits earned, course performance data, and admission and enrolment data?

- ○ Acceptable even if it does not improve identification
- ○ Acceptable only if there is at least a small improvement (about 1–3 additional students identified out of 100 students experiencing academic difficulties)
- ○ Acceptable only if there is at least a moderate improvement (about 5 additional students identified out of 100 students experiencing academic difficulties)
- ○ Acceptable only if there is at least a large improvement (about 10 additional students identified out of 100 students experiencing academic difficulties)
- ○ Never acceptable

Q3.5 **Data and privacy - sensitive data**
Some types of personal data are often considered particularly sensitive and raise additional privacy or ethical concerns.

In this question, each sensitive data type is considered as an **additional data source**, added on top of all non-sensitive data described earlier in this section.

For each type of sensitive data below, indicate the **minimum improvement** that would make it acceptable to include this data in the model as in the questions above. Again, an improvement means correctly identifying additional students who are actually experiencing academic difficulties, out of 100 such students, while keeping the number of students incorrectly flagged at about the same level.

A small improvement: about 1–3 additional students
A moderate improvement: about 5 additional students
A large improvement: about 10 additional students

	Acceptable if it does not worsen the model	Acceptable only if there is at least a small improvement	Acceptable only if there is at least a moderate improvement	Acceptable only if there is at least a large improvement	Never acceptable
Age	○	○	○	○	○
Gender	○	○	○	○	○
Nationality	○	○	○	○	○
Disability-related information	○	○	○	○	○
Race/ethnicity	○	○	○	○	○
First-generation college student status	○	○	○	○	○
Financial aid status	○	○	○	○	○

Q3.6 Sometimes, including sensitive personal data does not increase how many students experiencing academic difficulties are identified overall, but it may help improve the fairness of the model. In this context, improving fairness means reducing differences across student groups in how well the model identifies students experiencing academic difficulties or avoids unnecessarily flagging students who are not experiencing difficulties. For each type of sensitive data below, please indicate the minimum condition under which you would find its use acceptable in **situations where it does not increase the total number of students identified**.

	Acceptable even if it does not improve overall identification or fairness	Acceptable only if it improves fairness across student groups	Never acceptable	Not sure
Age	○	○	○	○
Gender	○	○	○	○
Nationality	○	○	○	○
Disability-related information	○	○	○	○
Race/ethnicity	○	○	○	○
First-generation college student status	○	○	○	○
Financial aid status	○	○	○	○

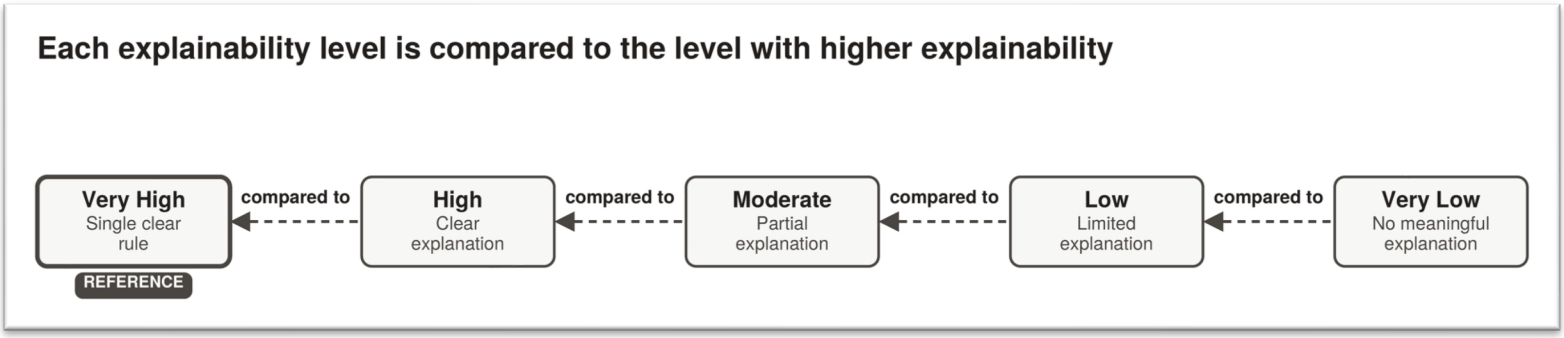
Q4.1 **Explainability:**
Predictive models for identifying students **who may be experiencing academic difficulties** differ in how clearly their decisions can be explained.

In this question, explainability refers to how well it is possible to explain **why a specific student was identified or not identified** as potentially experiencing academic difficulties.

There are competing perspectives:

- Some argue that explainability is essential. Advisors and students need to understand why someone was flagged to build trust and provide appropriate support, even if it means identifying fewer students who are experiencing academic difficulties.
- Others argue that effectiveness matters most. If a less explainable model identifies more students who are experiencing academic difficulties, that benefit outweighs the difficulty in explaining decisions. Both perspectives raise important questions. The following questions ask about your views on explainability trade-offs.

For each level of explainability below, indicate the minimum improvement that would make it acceptable to use a model with that level of explainability, compared to a higher level of explainability on this scale.



<u>Levels of explainability (with examples)</u>
Very high explainability (single clear rule) Each decision is based on one simple and transparent rule. Example: a student is identified if they have passed fewer than a certain number of course credits.

High explainability (clear explanation) Each decision can be explained using a small set of clear reasons with stable direction. Example: a student is identified because of a combination of low course credits, recent failed courses, and delayed progression, and it is possible to explain how each factor contributed.

Moderate explainability (partial but informative explanation) Each decision can be explained in general terms by highlighting the most influential factors, but the full reasoning cannot be reduced to a small, stable explanation.
Example: A student is identified based on many aspects of their study history, where it is possible to point to the most important contributing factors, but not fully explain the decision in a simple way.

Low explainability (limited, case-specific explanations) Only limited or approximate explanations are possible.
For a specific student, it may be possible to give a rough or partial explanation (for example, highlighting some contributing factors), but the explanation is incomplete and difficult to communicate clearly.

Very low explainability (no meaningful case-specific explanation) Decisions are produced by highly complex models whose internal reasoning cannot be meaningfully interpreted.
For a specific student, it is generally not possible to explain why they were identified or not identified, beyond stating that the model detected a pattern.

In the questions below, an improvement means correctly identifying additional students who are actually experiencing academic difficulties, out of 100 such students, while keeping the number of students incorrectly flagged at about the same level.

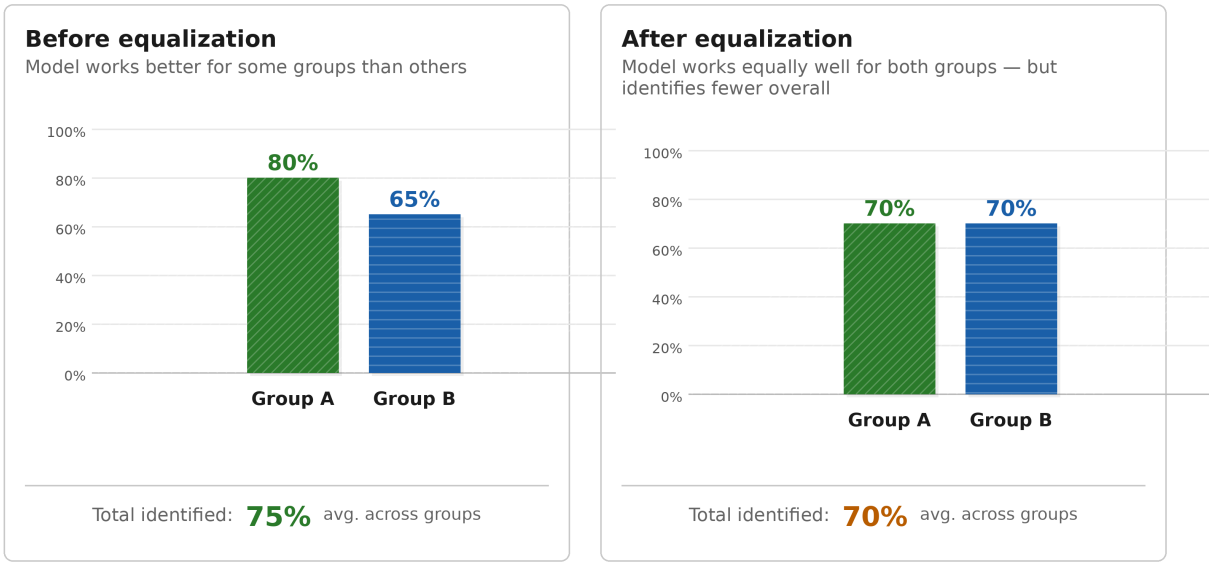
A small improvement: about 1–3 additional students
A moderate improvement: about 5 additional students
A large improvement: about 10 additional students

	Acceptable even if it does not improve identification	Acceptable only if there is at least a small improvement	Acceptable only if there is at least a moderate improvement	Acceptable only if there is at least a large improvement	Never acceptable
High explainability (few clearly defined factors)	○	○	○	○	○
Moderate explainability (many additive factors)	○	○	○	○	○

Low explainability (limited, case-specific explanations)	○	○	○	○	○
Very low explainability (no meaningful case-specific explanation)	○	○	○	○	○

Q5.1 **Fairness:**
When a predictive model is used to identify students who may be experiencing academic difficulties, it may not work equally well for all student groups — for example, identifying a higher share of students in one group than another (e.g. based on sex or nationality). The charts below show what this looks like, and what happens when the model is adjusted to close that gap.

> The charts show what happens when the model is adjusted to work equally well across two student groups.
> Before equalization: more students identified overall, but Group B is missed more often.
> After equalization: both groups treated equally — but fewer students reached in total.



As the charts show, making the model work equally well across groups may require accepting that fewer students are correctly identified overall.

There are competing perspectives:

- **Some argue** that overall effectiveness matters most. If more total students are helped, the model is doing its job even if it works better for some groups than others.
- **Others argue** that unequal performance perpetuates discrimination. A model that works worse for already-marginalized students deepens existing inequalities rather than helping.

Both perspectives raise important questions. The following questions ask about your views on fairness trade-offs.

How large of a decrease in correctly identifying students who are experiencing academic difficulties would you find acceptable in order to make the model work equally well across different student groups?

- ○ No decrease is acceptable, even if this means the model does not work equally well across student groups
- ○ About 1–3 fewer students identified overall out of 100 students experiencing academic difficulties
- ○ About 5 fewer students identified overall out of 100 students experiencing academic difficulties
- ○ About 10 fewer students identified overall out of 100 students experiencing academic difficulties
- ○ Not sure

Q5.2 When a predictive model is used to identify students who may be experiencing academic difficulties, it may make different types of errors for different student groups.

For example, the model may be more likely to miss students who need support in some groups, or more likely to unnecessarily contact students who do not need support in other groups. When deciding how to make the model fair across student groups, which of the following should be prioritized to be equal across groups?

- ○ Making the model equally good across student groups at identifying students experiencing academic difficulties who would otherwise be overlooked by the system
- ○ Making the model equally good across student groups at avoiding unnecessary contact with students who are not experiencing academic difficulties
- ○ Both goals should be prioritized equally
- ○ Not sure

Q6.1 **Intervention preferences:**
If a model indicates that a student may be experiencing academic difficulties, the university could respond in different ways. Which of the following interventions do you think would be most acceptable? **(Please select up to two.)**

- [ ] Receiving a confidential email or message with general information about available support services
- [ ] Receiving a confidential, personalized email or message that refers to the student's situation and suggests relevant support options
- [ ] An invitation to a voluntary meeting or call with a student advisor
- [ ] Receiving a confidential phone call from a student advisor to discuss support options
- [ ] No proactive outreach is taken; students receive support only if they explicitly request help
- [ ] Other (please fill in)

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Q6.2 How often should such outreach occur?

- ○ Once per year
- ○ Twice per year
- ○ Continuously — a student should be contacted as soon as the model identifies them as potentially experiencing academic difficulties, regardless of when that happens
- ○ Not sure

Q7.1 **Interventions and performance:**
When a system is used to identify students **who may be experiencing academic difficulties**, it has to decide how selective it should be. Assume that students identified by the system receive **a brief email with general information about available support services.**

There are two approaches:

- One approach is to contact many students, even if some of them turn out not to need support.
- Another approach is to contact fewer students, which reduces unnecessary contact but may miss some students experiencing academic difficulties.

Which approach do you think the university should take in this case?

- ○ Contact many students, even if some do not actually need support
- ○ Slightly prefer contacting many students, accepting that some students who do not need support may be contacted
- ○ Aim for a balance between contacting many students and contacting fewer students
- ○ Slightly prefer contacting fewer students, accepting that some students experiencing academic difficulties may be missed
- ○ Contact fewer students, even if some students experiencing academic difficulties are missed
- ○ Not sure

Q7.2 Now assume that students identified by the system receive **a personalized email or message that refers to the student's situation and suggests relevant support options** from a student advisor. In this case, which approach do you think the university should take?

- ○ Contact many students, even if some do not actually need support
- ○ Slightly prefer contacting many students, accepting that some students who do not need support may be contacted
- ○ Aim for a balance between contacting many students and contacting fewer students
- ○ Slightly prefer contacting fewer students, accepting that some students experiencing academic difficulties may be missed
- ○ Contact fewer students, even if some students experiencing academic difficulties are missed
- ○ Not sure

Q8.1 Is there any important concern, consideration, or perspective related to the use of predictive models for student support that you feel was not covered in this survey?

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Q9.1 When answering the questions in this survey, how often did you think about students you know who are having academic difficulties?

- ○ Not at all
- ○ Occasionally
- ○ About half the time
- ○ Often
- ○ Always
- ○ Prefer not to answer

ENDLINE survey

Q10.1 To what extent do you feel that the workshop improved your understanding of how predictive models could be used for student support?

- ○ Not at all
- ○ Slightly
- ○ Moderately
- ○ Substantially
- ○ Very substantially

Q10.2 Compared to before the workshop, how would you describe your overall view on using predictive models to support students?

- ○ I have become more supportive
- ○ I have become slightly more supportive
- ○ My view has not changed
- ○ I have become slightly more critical
- ○ I have become more critical

Q10.3 What most contributed to this shift? Select all that apply.

- [ ] Hearing a perspective from another participant I hadn't considered
- [ ] Learning about a specific trade-off (e.g., data use, explainability, fairness)
- [ ] Realizing the intervention design matters as much as the model
- [ ] A concrete example or scenario discussed
- [ ] The group's overall tone or apparent consensus
- [ ] Other (please fill in) _____

Q10.4 Why do you think your view did not change? Select the option that fits best.

- [ ] The workshop confirmed what I already believed
- [ ] I heard arguments on both sides that balanced out
- [ ] My view is based on values or principles that I don't think discussion can easily change
- [ ] The workshop did not introduce perspectives or information I hadn't already considered
- [ ] Other (please fill in) _____

Q10.5 During the workshop discussions, whose perspectives most influenced how you thought about the issues? (Select up to two.)

- [ ] Other students at my table
- [ ] The student advisor at my table
- [ ] Perspectives shared in the plenary discussion
- [ ] The facilitator
- [ ] I mainly reflected on the survey questions themselves

Q10.6 Was there anything during the workshop that influenced your view or made you realize something you had not previously considered about predictive models for student support? This could include reasons why your view did or did not change.

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